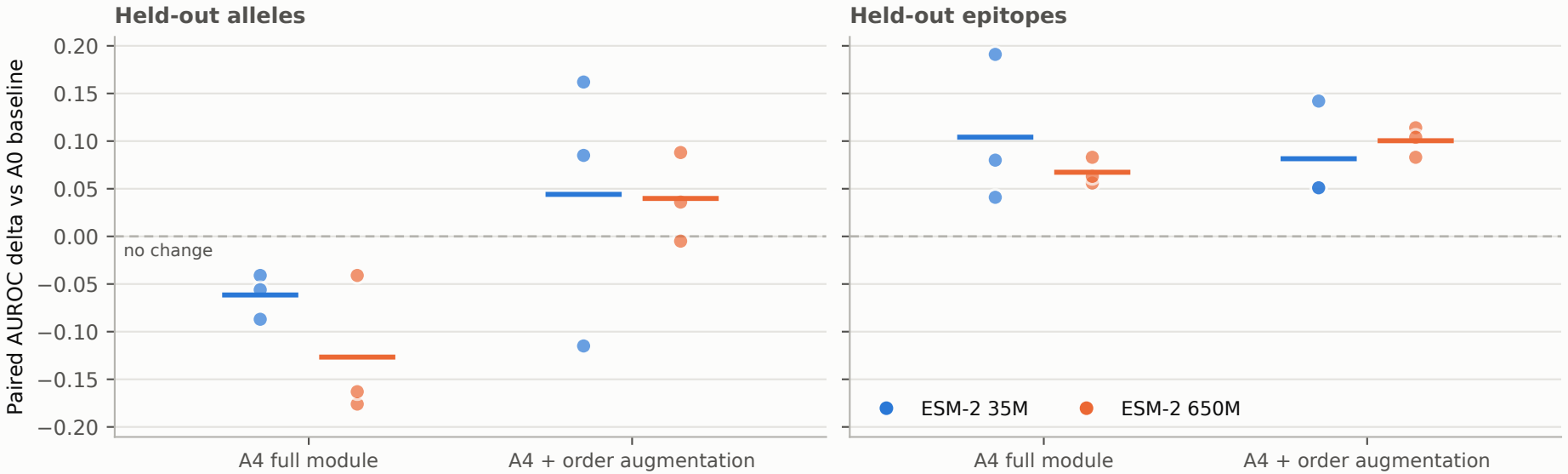
Paired by seed, the module raises held-out-epitope AUROC in every seed at both sizes

Each dot is one seed's delta against the baseline trained on the same subsample, initialization and batch order. The bar is the mean of the three.



Three contrasts agree in sign across all six seeds: A4 down on alleles, A4 and A5 up on epitopes. A sign test over six seeds gives p = 0.031 before correcting for the four contrasts shown, and nothing survives that correction. The baseline is below chance on held-out epitopes, so an increase there moves toward 0.5 rather than above it. This pairing was chosen after seeing the unpaired result.